

# AI Agents for Data Visualization and Kaggle Competitions

An end-to-end LLM agent (my-agent) benchmarked against two frozen yardsticks on 20 Kaggle competitions — with the apparatus needed to make the comparison honest

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## 1 · Motivation

### Why these tasks are hard

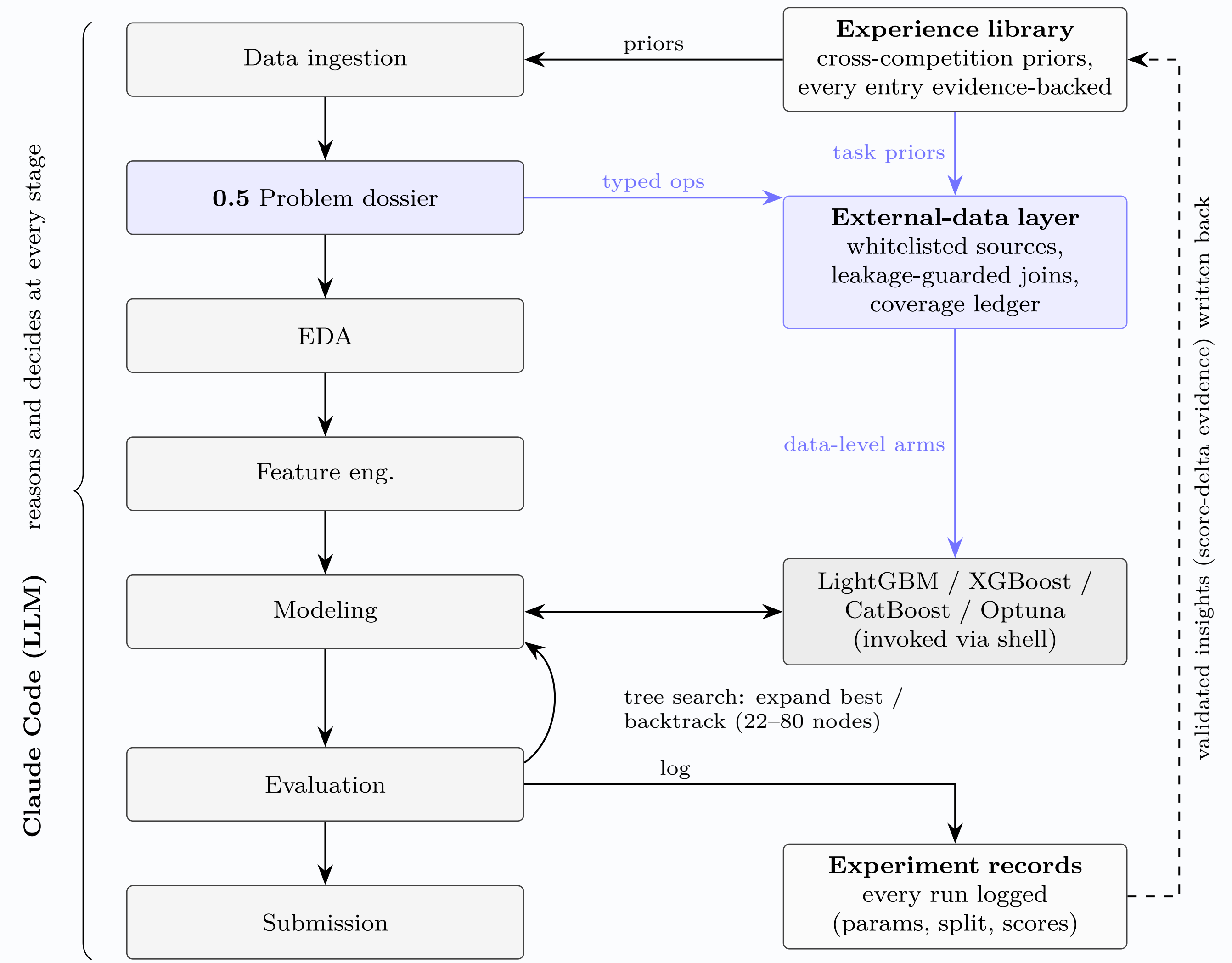
- **Kaggle competitions** demand the full expert loop — validation design, feature engineering, model selection, ensembling — where the difference between top entries sits at the **4th–5th decimal** of the metric.
- With gaps that small, **local cross-validation is an unreliable compass**: a pipeline’s own validation score routinely disagrees with the real leaderboard, so no local number can be trusted to pick a winner.
- **Data visualization** (EDA, diagnostics, reporting) is essential but repetitive: every dataset needs distribution, drift, and correlation views before modeling can be trusted.

### Why an AI agent

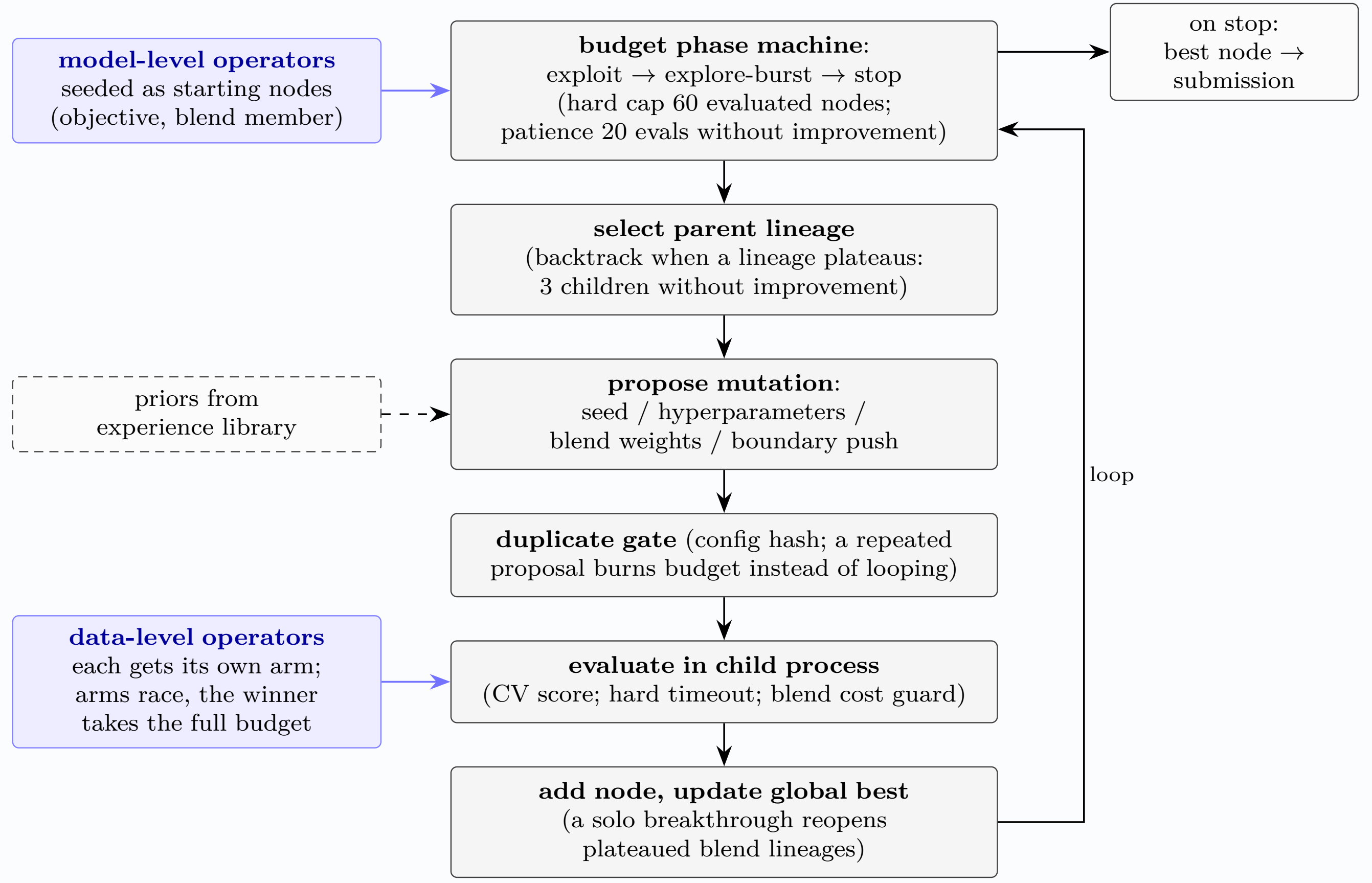
- Tireless, systematic search: 22–80 candidate configurations per competition, every experiment logged and reproducible.
- Runs the same disciplined protocol on 20 competitions serially — a human cannot hold this consistency.
- Generates per-dataset EDA figures and per-competition reports automatically, so every claim is grounded in an artifact.

## 2 · Workflow

An LLM reasons at every stage; Auto-ML libraries are invoked through the shell; a cross-competition **experience library** supplies evidence-backed priors; a **tree search** over candidate configurations is the optimisation loop.



Six-stage pipeline: problem dossier → EDA → features → modeling → evaluation → submission. The v5 dossier layer classifies the task and injects external data *upstream*, before any modeling.



Tree-search loop: ensembles are first-class nodes with OOF caching and weight search; exploit until plateau, then one forced exploration burst.

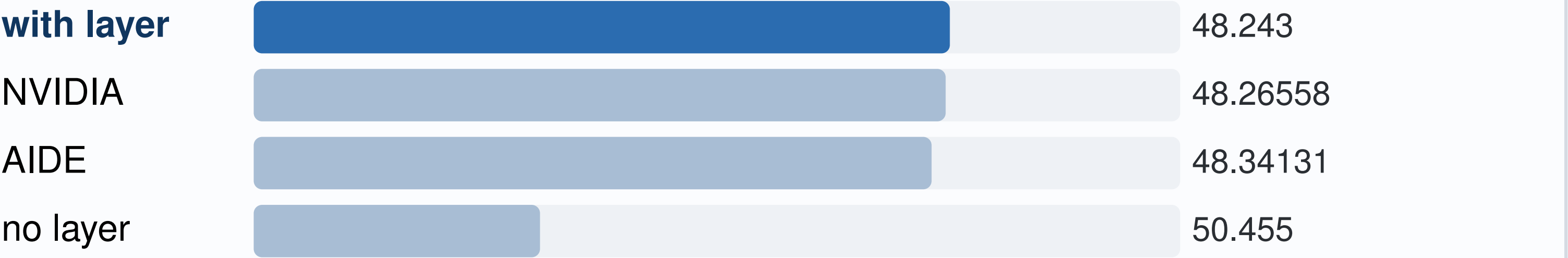
## 3 · Results — Kaggle benchmark

All submissions scored on the real leaderboard (late submission); verdicts adjudicated by a paired test built from Kaggle’s own public/private split.

### Benchmark standing (scored 2026-08-18, 16 of 20 lanes)

All 20 competitions run under the frozen Stage-0.5 architecture in credential-free sandboxes with a per-lane transcript audit. The 16 completed lanes are scored on the real leaderboard, each duel adjudicated by a paired test built from Kaggle’s own public/private split: **my-agent leads the standing — 15 wins / 5 losses (75% excl. ties)** vs. AIDE 42% and NVIDIA 38% — and takes the **best private score on 10 of 16**. One lane running (s6e2), three queued (**TBD**).

### Injection-layer ablation (s3e19, private SMAPE ↓)



The dossier layer’s external-data injection turns the architecture’s canonical failure into a numerical first among the three agents — and all **6 of 6** controlled external-data arms beat their matched baselines. Bar length  $\propto$  margin over 52 (lower SMAPE is better).

### Mid-complexity three-way (offline-graded, private split)

| Competition              | Metric        | my-agent       | NVIDIA  | AIDE    |
|--------------------------|---------------|----------------|---------|---------|
| us-patent (NLP)          | Pearson $r$ ↑ | <b>0.86658</b> | 0.86160 | 0.65821 |
| ventilator (time series) | MAE ↓         | <b>0.16525</b> | 0.40093 | 0.34591 |
| SIIM-ISIC (imaging)      | AUC ↑         | <b>0.93529</b> | 0.89822 | 0.88579 |

## 4 · Advantages & Difficulties

### Expected → experienced

- **Expected:** the agent would quickly beat most humans. **Experienced:** development runs landed above average, with wins concentrated where no mature public solutions exist; the frozen-architecture re-run now leads the paired standing on the 16 scored lanes.
- **Expected:** one run and one score decide a winner. **Experienced:** score noise and run-to-run variance leave many duels undecidable; a paired error model became mandatory.
- **Expected:** the agent follows the rules by default. **Experienced:** a transcript audit caught one lane reading its own competition’s recorded findings — and the clean re-run scored *worse*, proving the contamination advantage was real.
- Fairness is expensive: isolated data roots, credential-free sandboxes, and per-lane transcript audits cost real engineering time — but without them the numbers are not comparable.

## 5 · Discussion — advice

- **State the error model with every claim.** A ranking without an error scale is noise: many of our duels are undecidable, and the winner flips when the evaluation set changes.
- **Never trust local CV for between-agent claims.** Score on the real leaderboard; CV describes a pipeline’s internal view only.
- **Freeze your yardsticks.** Improving a baseline mid-study silently turns the benchmark into an optimisation of the baseline.
- **Isolate mechanically, then audit the transcript.** Sandboxes stop leaks you anticipated; reading what the agent actually did catches the ones you didn’t — in both directions (we hit one real contamination and one auditor false positive).
- **Put knowledge injection upstream.** Injected at the blend stage it was worth  $\sim 10^{-5}$ ; moved into problem identification it lifted the canonical failure case from last to first.

### Where knowledge enters

- Downstream (blend stage): measured gain  $\sim 10^{-5}$  — abandoned.
- Upstream (problem dossier): fires on exactly the **2/18** competitions that need external data; all **6 of 6** controlled external-data arms beat their matched baselines.
- But no local signal can select the *form* of injection — the one surviving selector died on its first pre-registered prediction.

Companion documents: Report v6 (Overleaf, main document) · Engineering Log · Case Studies · 22 per-competition ML-spec reports.